IPL Next Season Performance Predictor

Machine Learning-Based Sports Analytics Solution

Presented By: Uday Deshmukh · AIML Student

PYTHON | PANDAS | SCIKIT-LEARN | STREAMLIT | GITHUB

The Business Problem

IPL franchises invest millions in player retention and auction decisions — yet most choices rely on intuition rather than data. The cost of a wrong pick is enormous.

Identifying Future Performers

No systematic way to spot breakout talent before the next season.

Decision Bias

Human judgment introduces inconsistency and recency bias in evaluations.

Underutilized Data

Historical performance data exists but is rarely leveraged predictively.

Team-Building Strategy

Franchises lack a data-driven framework for squad composition.

Project Objectives

This project delivers an end-to-end ML solution — from raw historical data to an interactive prediction dashboard — empowering franchises to make smarter, evidence-based decisions.

1

Analyze Historical Data

Process multi-season IPL player statistics to establish performance baselines.

2

Engineer Features

Create derived metrics like Batting Impact Score and Runs Per Match for richer modeling.

3

Train ML Model

Build and validate a binary classification model to predict High Performer status.

4

Deploy Dashboard

Launch an interactive Streamlit app for real-time player evaluation and prediction.

Dataset & Features

Raw Performance Features

- Runs & Balls Faced
- Batting Average & Strike Rate
- Economy Rate & Wickets Per Match
- Matches Played

Engineered Features

Runs Per Match

Normalizes output across varying match counts.

Balls Per Match

Measures opportunity and consistency.

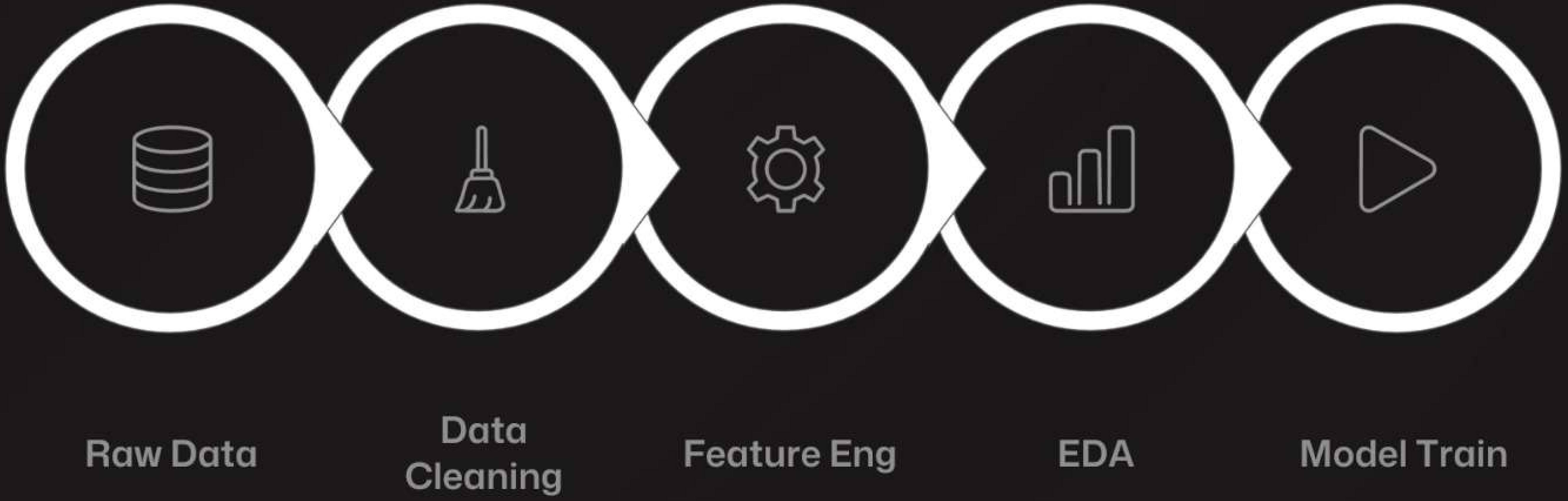
Batting Impact Score

Composite metric combining SR and average.

Target Variable: High Performer Next Season — Yes / No

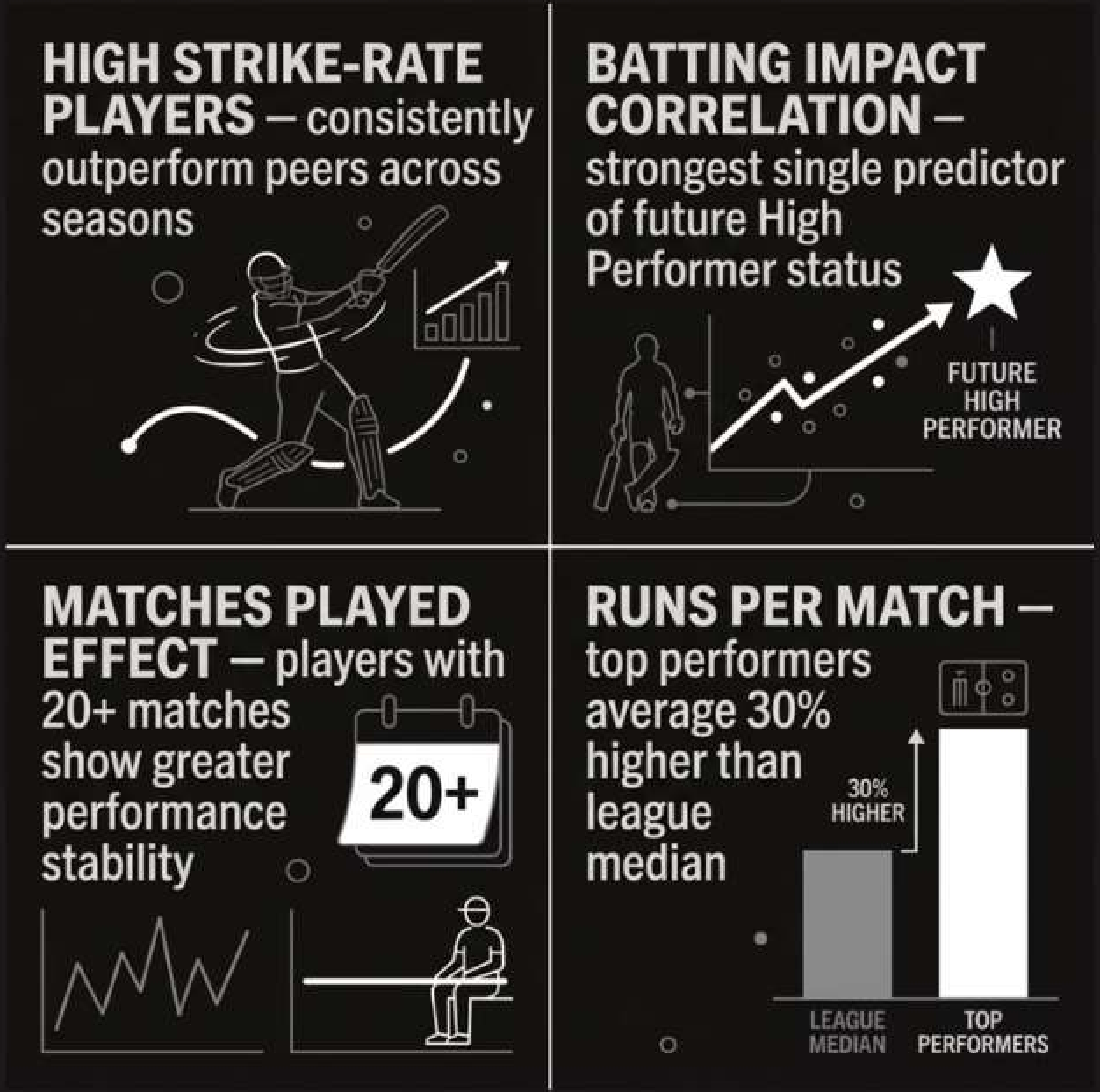
Data Processing Pipeline

A structured, reproducible ML workflow transforms raw IPL statistics into actionable performance predictions — ensuring data quality at every stage before modeling.



Each stage includes rigorous quality checks — from handling missing values and outliers to label creation and data transformation — ensuring the model trains on clean, meaningful signals.

Exploratory Data Analysis



What the Data Revealed

EDA surfaced clear performance signals before any model was trained. Strike rate and batting impact emerged as the most powerful differentiators between future high performers and the rest of the league.

Strike rate distribution showed a right-skewed pattern favoring elite performers.

Correlation heatmap confirmed Batting Impact Score as the top feature.

Experience (matches played) correlated positively with consistency.

Machine Learning Model

LOGISTIC REGRESSION · BINARY CLASSIFICATION

Logistic Regression was selected for its interpretability, speed, and strong baseline performance on tabular sports data — making it ideal for stakeholder-facing analytics.

Fast & Efficient

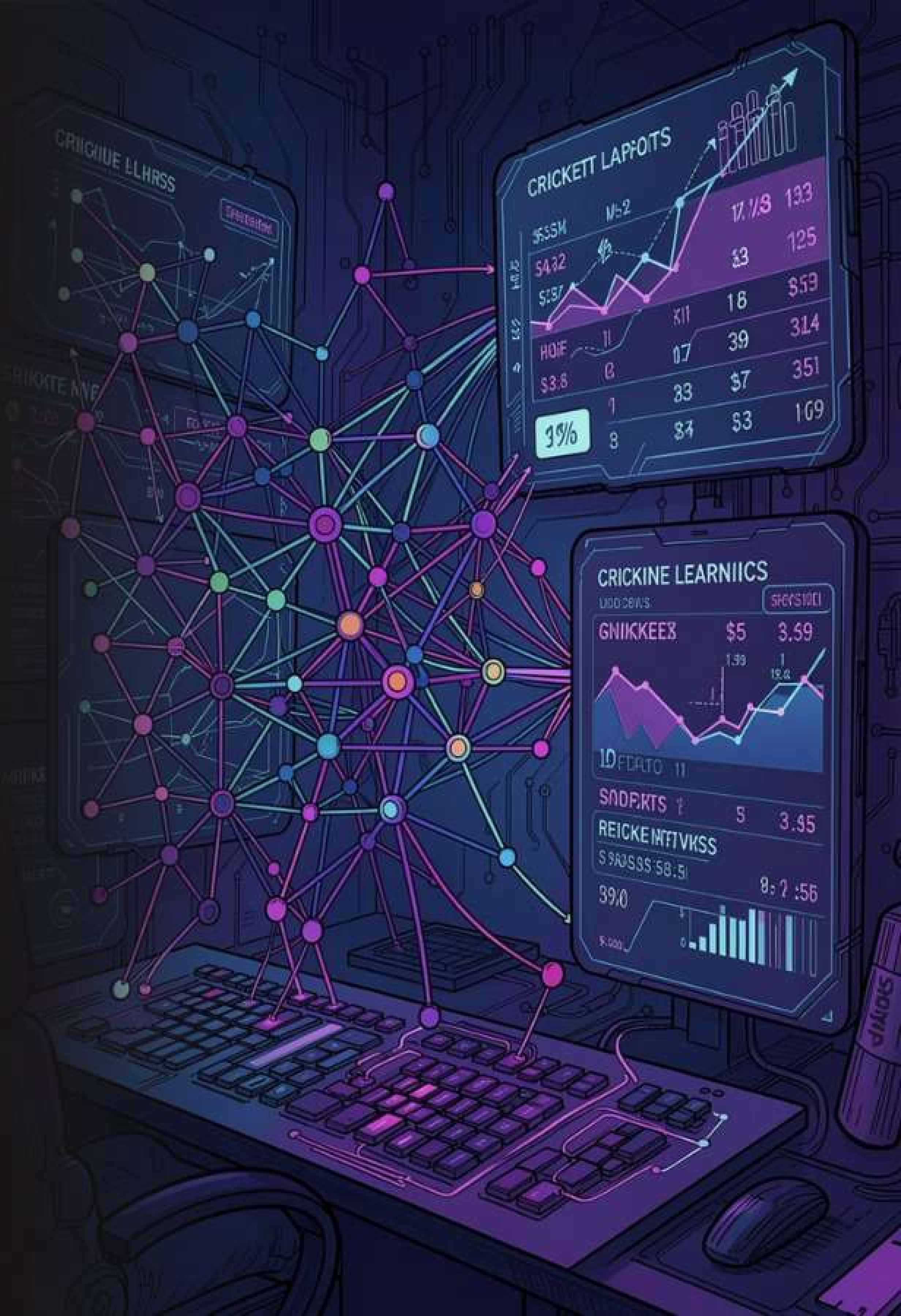
Minimal compute overhead — suitable for real-time dashboard predictions.

Fully Interpretable

Feature coefficients reveal exactly which stats drive each prediction.

Binary Output

Naturally maps to the High Performer Yes/No classification task.



Model Performance

85%	59%	55%
Accuracy	**Precision**	**Recall**
Correctly classifies 85 out of 100 players overall.	Of players flagged as High Performers, 59% truly are.	Captures 55% of all actual future high performers.

What This Means for Franchises

The model provides a reliable analytical foundation for shortlisting players. At 85% accuracy, it significantly outperforms random or intuition-based selection, reducing the risk of costly auction mistakes.

Interpretation: Precision and Recall can be tuned based on franchise risk appetite — prioritize recall to avoid missing talent, or precision to minimize false positives.

Interactive Streamlit Dashboard

Built for Decision-Makers

The Streamlit dashboard translates model outputs into an intuitive, no-code interface — accessible to analysts, scouts, and franchise owners alike.

- **Player Selection** — Dropdown with full-name search across all IPL players.
- **Performance Prediction** — Instant High Performer Yes/No output with confidence score.
- **Top 10 Leaderboard** — Ranked list of highest-probability future performers.
- **Downloadable Results** — Export predictions for stakeholder reports.

The Business Problem

IPL franchises invest millions in player retention and auction decisions — yet most choices rely on intuition rather than data. The cost of a wrong pick is enormous.

Identifying Future Performers

No systematic way to spot breakout talent before the next season.

Decision Bias

Human judgment introduces inconsistency and recency bias in evaluations.

Underutilized Data

Historical performance data exists but is rarely leveraged predictively.

Team-Building Strategy

Franchises lack a data-driven framework for squad composition.